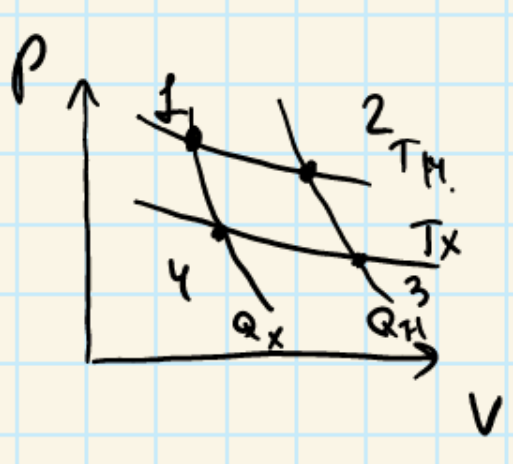
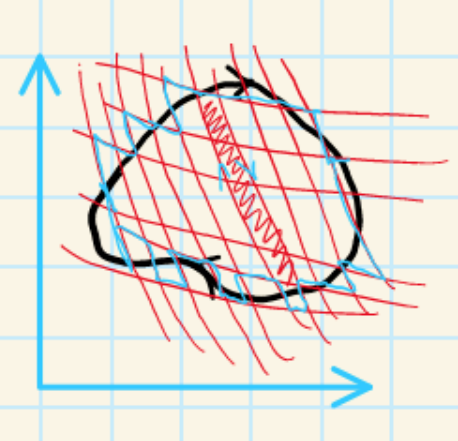
# Доп семинар (18фев)

18 февраля 2026 г. 18:36



$$\frac{Q_н}{T_н} = \frac{|Q_х|}{T_х}$$



$N_{циклов}$

$T_1, T_2, \dots, T_N$ → 0

$Q_1, Q_2 \dots Q_n$

$T_1' \dots T_n'$

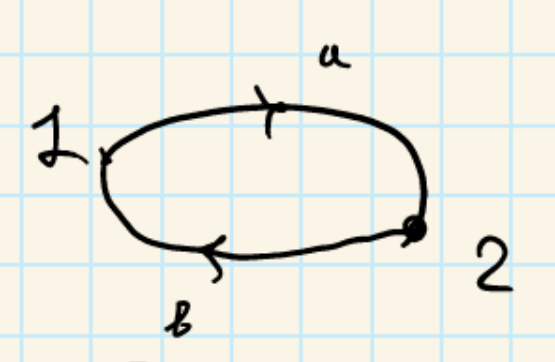
$Q_1' \dots Q_n'$

$$\frac{Q_i}{T_i} = \frac{|Q_i'|}{T_i'}$$

$$\sum_{i=1}^{N}\left(\frac{Q_i}{T_i} - \frac{|Q_i'|}{T_i'}\right) = 0$$ это стремится к интегралу

$$\underbrace{\oint \frac{\delta Q}{T}}_{\text{энтропийный интеграл}} = 0$$

энтропия $dS$



$$\oint \frac{\delta Q}{T} = \int_{1a2} \frac{\delta Q}{T} + \int_{2b1} \frac{\delta Q}{T} = 0$$

Меняем течение процесса

$$\int_{1a2} \frac{\delta Q}{T} = \int_{2b1} \frac{\delta Q}{T}$$

$$\delta Q = T\,dS = \overset{C_v dT}{dU} + p\,dV$$

$$dS = C_v \frac{dT}{T} + R\frac{dV}{V} \quad (1 \text{ моль})$$

$$S(T,V) = \nu\left(C_v \ln T + R \ln \frac{V}{\nu} + S_0\right)$$

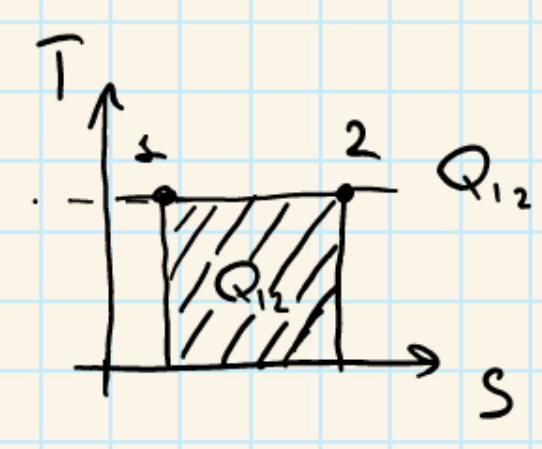
молярный объём | константа

Наизусть !!!

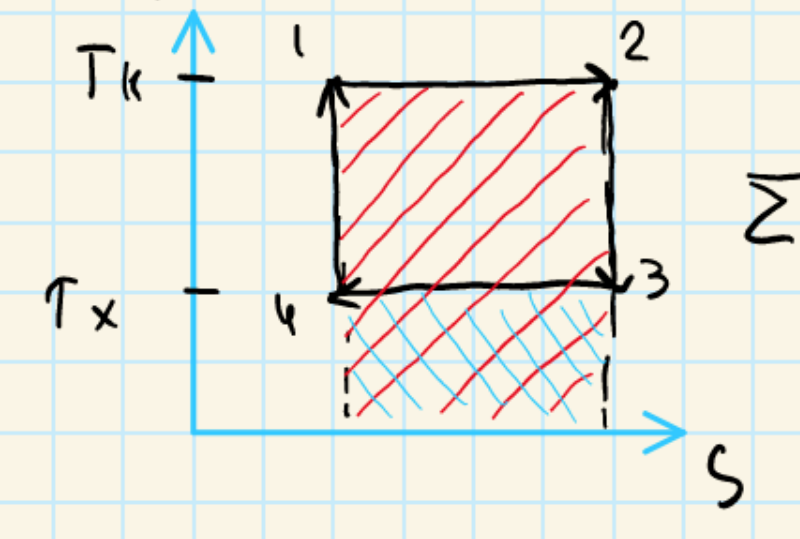
$$S(T,P) = \nu\left(C_p \ln T - R \ln P + S_0'\right)$$

константа

$$\delta Q = T\,dS$$



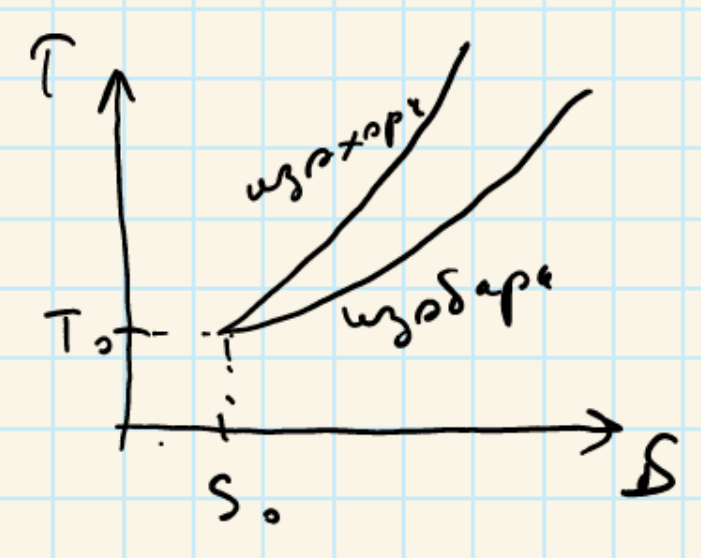
$$Q_{12} = T(S_2 - S_1)$$



$$\sum Q_i = \sum A_i = A$$

Политропа $\delta Q = C\,dT = T\,dS$

$$\int_{T_0}^{T} \frac{dT}{T} = \int_{S_0}^{S}$$



$$\ln\frac{T}{T_0} = \frac{S - S_0}{C}$$

$$T(S) = \exp\left(\frac{S - S_0}{C}\right)$$

## Т 2.1.

Т.Н.

1моль $T_0 = 300K$ → до $T_1 = 200K$

М

1 моль $T_0 = 300K$ $T_2 = ?$

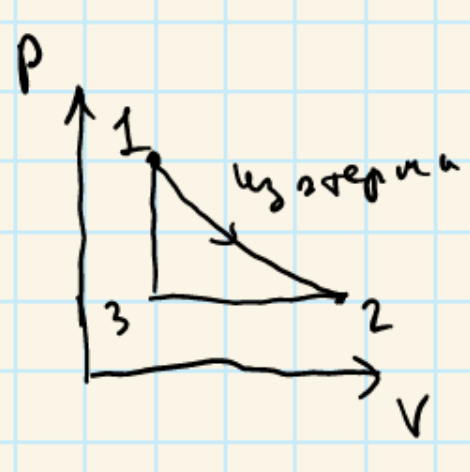
1) $\oint \frac{\delta Q}{T} = 0 \longrightarrow \oint C_v \frac{dT}{T} \longrightarrow \int_{T_0}^{T_1} C_v \frac{dT}{T} + \int_{T_0}^{T_2} C_v \frac{dT}{T} = 0$

$$\ln\frac{T_1}{T_0} + \ln\frac{T_2}{T_0} = 0 \Rightarrow \ln\frac{T_1 T_2}{T_0^2} = 0$$

$$T_2 = \frac{T_0^2}{T_1} = \frac{300 \cdot 300}{200} = 450K$$

$$A_{min} = U_{нач} - U_{кон} = C_v \cdot 2T_0 - C_v T_1 - C_v T_2 =$$
$$= C_v(2 \cdot 300 - 200 - 450) = -1040 Дж.$$

## №4.16 (Близко к Д/З)

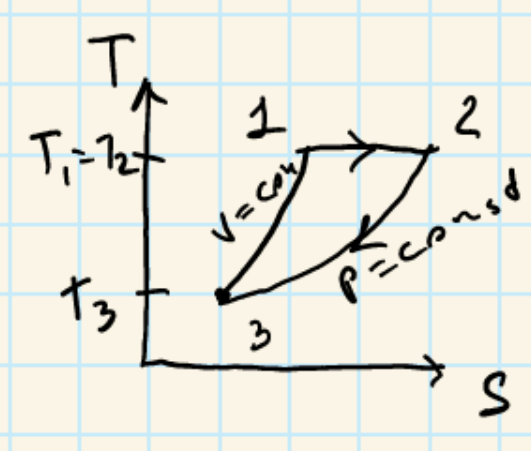


$\eta_{1231}$ − ?

Рабочее вещ-во идеал.

$$\frac{C_p}{C_v} = \frac{4}{3}$$

$$a = \frac{T_1}{T_3} = 2$$



$$\eta = \frac{\sum Q_i}{Q^+} = \frac{Q_{12} + \overset{<0}{Q_{23}} + Q_{31}}{Q_{12} + Q_{31}}$$

$$Q_{12} = T_1(S_2 - S_1);\quad Q_{23} = C_p(T_3 - T_2);$$
$$Q_{31} = C_v(T_1 - T_3)$$

$$\oint \frac{\delta Q}{T} = 0$$

$$S_3 - S_2 = C_p \ln\frac{T_3}{T_2}$$
$$+\; S_1 - S_3 = C_v \ln\frac{T_1}{T_3}$$

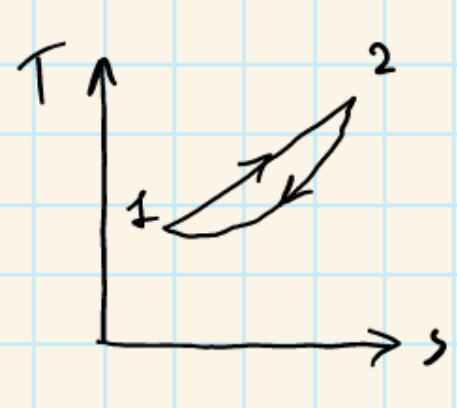
$$S_1 - S_2 = C_p \ln\frac{T_3}{T_2} + C_v \ln\frac{T_1}{T_3}$$

$$S_2 - S_1 = C_p \ln\frac{T_2}{T_3} - C_v \ln\frac{T_1}{T_3}$$

$$Q_{12} = T_1(S_2 - S_1) = T_1\left(C_p \ln\frac{T_2}{T_3} - C_v \ln\frac{T_1}{T_3}\right) =$$
$$= |T_1 = T_2| = T_1 \ln \overset{T_1/T_3}{a}\,(C_p - C_v)$$

$$\eta = 1 + \frac{Q_{23}}{Q_{12} + Q_{23}} = 1 - \frac{\gamma(a-1)}{a \ln a(\gamma - 1) + a - 1} = 0{,}088$$

## №4.22



$C_I$ − ?

$\eta = 0{,}2;\quad C_{II}^{max} = 45 \frac{Дж}{К \cdot моль}$

$C_{II}^{min} = 35 \frac{Дж}{К \cdot моль}.$

Ⓘ $T = T_0 e^{\frac{S - S_0}{C}}$

ⅠⅠ $S = S_0 + \alpha T$

$$\delta Q_{II} = T\,dS_{II} = C(T)\,dT$$

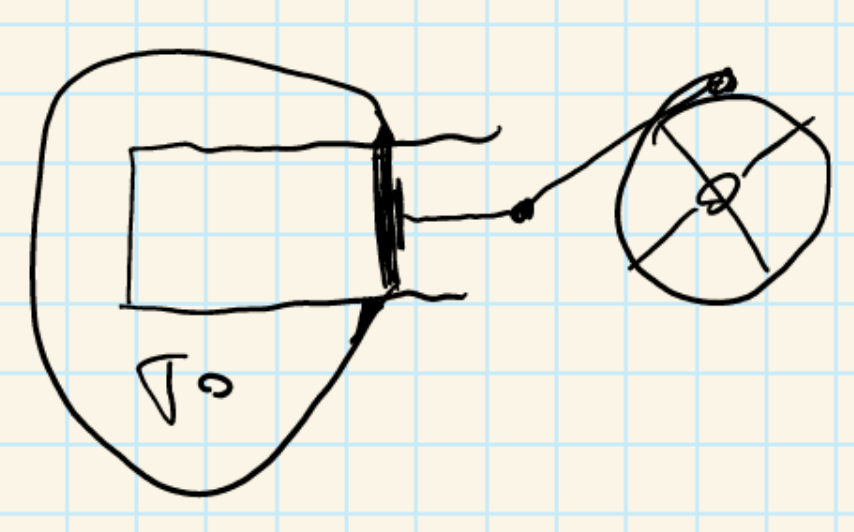
$$\alpha = \left(\frac{\partial S}{\partial T}\right)_{II} = \frac{C(T)}{T}$$

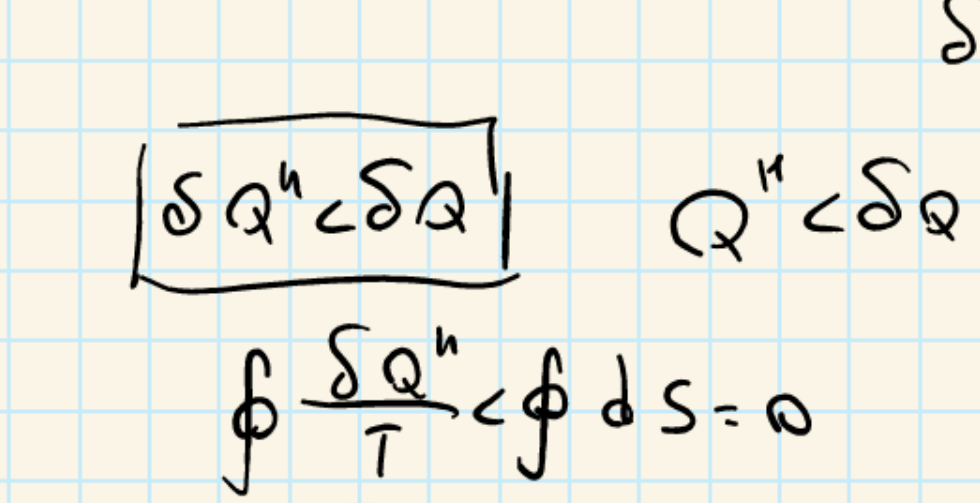
$$Q_I = C(T_1 - T_2) < 0$$

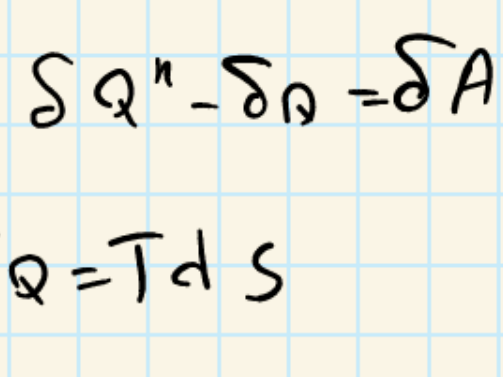
$$Q_{II} = \int_{T_1}^{T_2} C(T)\,dT = \int_{T_1}^{T_2} \alpha T\,dT = \frac{\alpha}{2}(T_2^2 - T_1^2) > 0$$

$$\eta = 0{,}2 = 1 - \frac{C(T_2 - T_1)}{\frac{\alpha}{2}(T_2^2 - T_1^2)} = 1 - \frac{2C}{\alpha(T_2 + T_1)}$$

$$C = \frac{1-\eta}{2}\left(C_{II}^{max} + C_{II}^{min}\right) = 32 \frac{Дж}{К \cdot моль}$$







$$dU = \delta Q^н - \delta A^н$$
$$dU = \delta Q - \delta A$$

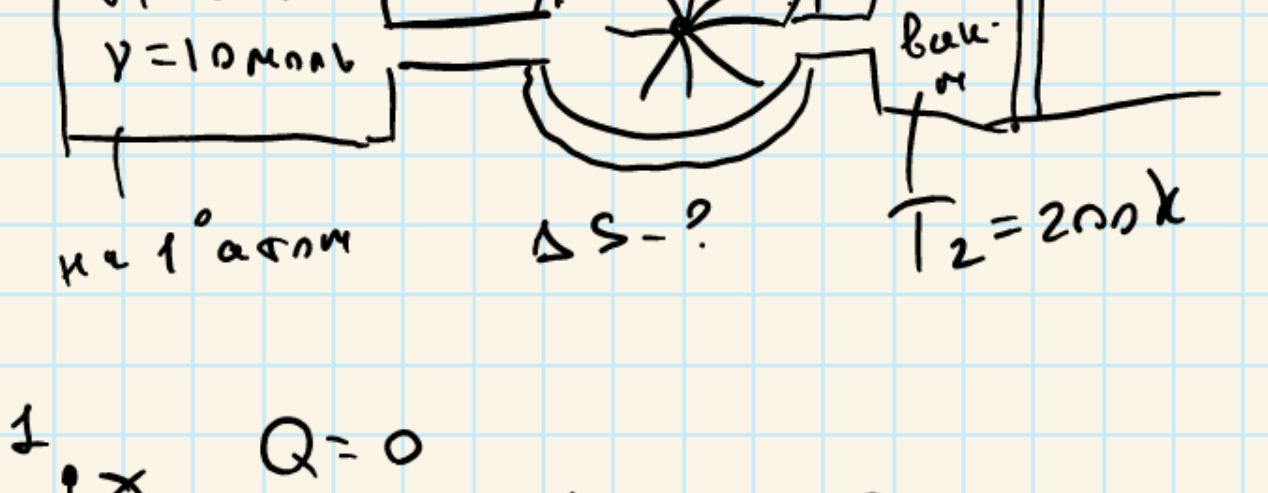
$$\delta Q^н - \delta Q = \delta A^н - \delta A < 0$$

$$\boxed{\delta Q^н < \delta Q} \qquad Q^н < \delta Q = T\,dS$$

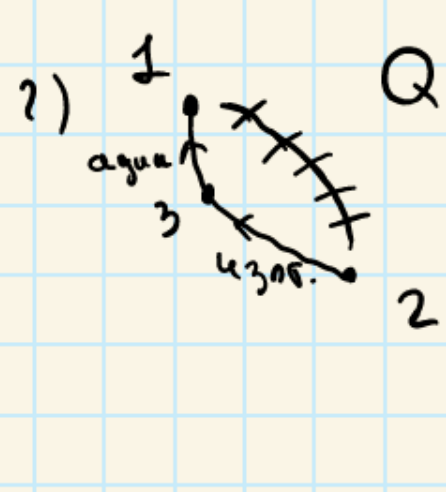
$$\oint \frac{\delta Q^н}{T} < \oint dS = 0$$

$$\boxed{\oint \frac{\delta Q^н}{T} < 0}$$

## №4.47 (Д/З)



2)



$Q = 0$

$$A = A_{231} = 15 Дж$$

$$\Delta S_{1-2}^{необ} = \Delta S_{132}$$

3) 1 − 3 адиабата. $Q = 0 = \nu C_v(T_3 - T_1) + A_{13}$

$$A_{13} = \nu C_v(T_1 - \underset{''T_2}{T_2})$$

4) 3 − 2 − изотерма $\Delta U = 0 \quad Q_{32} = A_{32}$

$$A_{32} = A - A_{13} = A - \nu C_v(T_1 - T_2)$$

$$\Delta S_{132} = \frac{Q_{32}}{T_2} = \frac{A}{T_2} - \nu C_v \frac{T_1 - T_2}{T_2} = 12{,}65 Дж$$